

Numerical Integration

① Closed Cotes

→ Trapezoidal rule $n=1$

$$a=x_0 \int_{x_0}^{x_1=b} f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)] \quad , \quad h = x_1 - x_0$$

→ Simpson's rule $n=2$

$$a=x_0 \int_{x_0}^{x_2=b} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \quad , \quad h = \frac{b-a}{2}$$

$\uparrow \frac{a+b}{2}$

→ Simpson's 3rd/8 rule $n=3$

$$a=x_0 \int_{x_0}^{x_3=b} f(x) dx = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] \quad , \quad h = \frac{b-a}{2}$$

→ $n=4$ ('ll be given, and asked in Q use $n=4$ formula)

$$\int_{x_0}^{x_4} f(x) dx = \frac{2h}{45} [7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)] \quad , \quad h = \frac{b-a}{2}$$

Q1 1. Approximate the following integrals using the Trapezoidal rule.

h. $\int_{x_0=0}^{x_1=\pi/4} \frac{e^{3x} \sin 2x}{f(x)} dx$

Trapezoidal rule $\int_{x_0}^{x_1} f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)]$

$h = x_1 - x_0 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$

$$\int_0^{\pi/4} e^{3x} \sin 2x dx = \frac{\pi}{4} [e^{3(0)} \sin 2(0) + e^{3(\pi/4)} \sin 2(\pi/4)]$$

 $= 4.143259655$

a. $\int_{0.5}^1 x^4 dx$ $\int_{x_0}^{x_1} f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)]$

$\int_{0.5}^1 x^4 dx = \frac{0.5}{2} [(0.5)^4 + (1)^4] = 0.265625$ $h = 1 - 0.5 = 0.5$

Q5 5. Repeat Exercise 1 using Simpson's rule. $\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$

c. $\int_1^{1.5} x^2 \ln x dx$ $h = \frac{1.5 - 1}{2} = 0.25$

$\int_{x_0=1}^{x_2=1.5} x^2 \ln x dx = \frac{0.25}{3} [f(1) + 4f(1.25) + f(1.5)] = 0.19224530$
 $x_1 = \frac{1 + 1.5}{2} = 1.25$

② Open Cotes

"in Close Cote the Value of 'n' will be given in Q"

* → Midpoint Rule ($n=0$) (NOT given)

$$\int_{x_{-1}}^{x_1} f(x) dx = 2h f(x_0) \quad h = \frac{b-a}{n+2}$$

* → $n=1$ (will be given)

$$\int_{x_{-1}}^{x_2=x_{n+1}} f(x) dx = \frac{3h}{2} [f(x_0) + f(x_2)] \quad h = \frac{b-a}{n+2}$$

* → $n=2$

$$\int_{x_{-1}}^{x_3=x_{n+1}} f(x) dx = \frac{4h}{3} [2f(x_0) - f(x_1) + 2f(x_2)] \quad h = \frac{b-a}{n+2}$$

* → $n=3$

$$\int_{x_{-1}}^{x_4=x_{n+1}} f(x) dx = \frac{5h}{24} [11f(x_0) + f(x_1) + f(x_2) + 11f(x_3)] \quad h = \frac{b-a}{n+2}$$

Q9

9. Repeat Exercise 1 using the Midpoint rule.

$$c. \int_1^{1.5} x^2 \ln x \, dx \quad h = \frac{b-a}{n+2} = \frac{1.5-1}{0+2} = 0.25$$

$$\int_{x_{i-1}}^{x_i} x^2 \ln x \, dx = 2h f(x_0)$$

$$\int_1^{1.5} x^2 \ln x \, dx = 2(0.25) \left((1.25)^2 \ln(1.25) \right) = 0.1743308$$

$$x_0 = \frac{x_1 + x_{i-1}}{2} = 1.25$$

Q13

13. The Trapezoidal rule applied to $\int_0^2 f(x) \, dx$ gives the value 4, and Simpson's rule gives the value 2. What is $f(1)$?

$$\text{Trapezoidal} \Rightarrow \int_0^2 f(x) \, dx = \frac{2}{2} [f(0) + f(2)] = 4 \quad h = 2 - 0 = 2$$

$$\Rightarrow f(0) + f(2) = 4$$

$$\text{Simpson's} \Rightarrow \int_0^2 f(x) \, dx = \frac{1}{3} [f(0) + 4f(1) + f(2)] = 2 \quad h = \frac{2-0}{2} = 1$$

$$\Rightarrow f(0) + 4f(1) + f(2) = 6$$

$$\Rightarrow \cancel{f(0)} + 4f(1) + \cancel{f(2)} = 6$$

$$\cancel{f(0)} + 0 + \cancel{f(2)} = 4$$

$$\frac{4f(1)}{4} = \frac{2}{4} = \frac{1}{2}$$

$$\Rightarrow \boxed{f(1) = \frac{1}{2}}$$